

GR VertiQ
SOLUTION SARL

THE FUTURE OF
VERTICAL MOBILITY



About Us



At GR VertiQ, we redefine urban mobility. We are a premier provider of advanced elevators and escalators solutions, dedicated to keeping this rapid world moving smoothly, safely, and efficiently. From architectural marvels to residential spaces, we are engineering seamless vertical transport systems that blend cutting-edge technology with elegant design.

we have Supply, installed, and maintained high-performance elevators and escalators for commercial, residential, and industrial sectors.

Core Strengths



All kinds of passenger elevators



High-speed passenger elevators



Rapid-response service and maintenance team



Eco-friendly public transit escalators

GR **VertiQ**
SOLUTION SARL

Elevator Ride



Safe and Reliable Travel.

Automatic emergency levelling device in case of power failure

When a power off during running, the automatic emergency leveling device can run the car to the nearest level position in a timely manner, so that passengers can safely leave the elevator.

Permanent magnet synchronous gearless traction machine, Reliable and durable

The power capacity is larger, the temperature rise is better than the national standard.

It provides 2.5 times the rated load braking torque, and a higher transmission efficiency.

The brake is tested by 15 million times.

Microcomputer control system, Stable operation

The new generation of dual 32-bit control system adopts CANBUS communication, direct docking, distance control and other technologies, ensuring stable operation, safety and reliability.

Permanent magnet door machine, Rapid and steady

The torque is greater, the efficiency is higher, and the operation stability and sensitivity are better.

The curtain light can make rapid response to any person or object when entering into the detection area.

The door system (door lock+ door machine) has passed 12 million times reliability tests.

Technology of preventing car's unintended movement, Safe leveling

The precision of car leveling detection can be up to millimeter without mistake, so that the passengers easily get in and out. The reliable car leveling memory technology ensures that the elevator stops in advance and the leveling is safe.

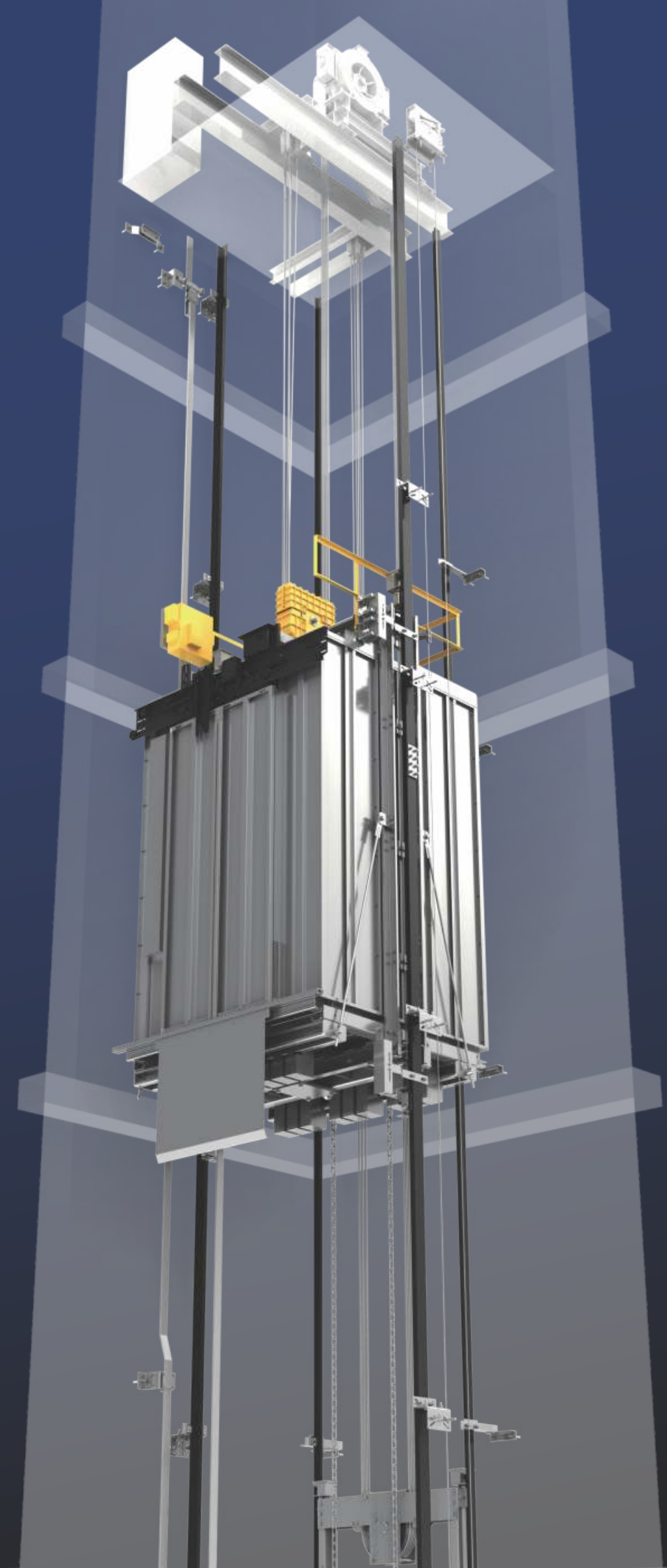
Security Components, Full Protection

The safety gear adopts symmetrical wedge block to avoid the occurrence of the car shaking while using unilateral wedge braking, and reduce the damage to the guide rail.

Full load and full speed hydraulic buffer with longer service life.

Car positioning technology, Safe arrival

The abnormal sliding between the wire rope and the traction wheel is effectively eliminated, and the possibility of returning to the home station by accident is reduced to a minimum.





Landing door series

Various type of landing doors apply to all various needs of buildings



Four-panel center opening
Landing
Dusting steel plate
Jamb
Dusting steel plate

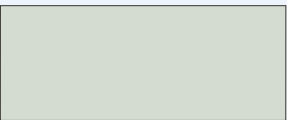


Side opening
Landing
Dusting steel plate
Jamb
Dusting steel plate

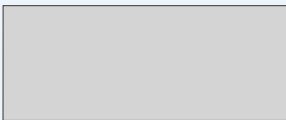


Six-panel center opening
Landing
Dusting steel plate
Jamb
Dusting steel plate

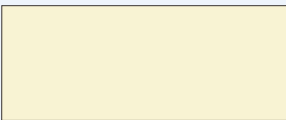
Dusting Color Standard (optional)



KL-SK 001
Ice gray



KL-SK 002
Plane gray



KL-SK 003
Comy yellow

Cabin and Landing Operation Panel
Optional screen / call indicator without bottom box



KLII3-4C
Standard

LED segment code



KLII3-4C
Standard



KL8II-IH
Standard



KL8I3-IH
Charged option

white dot matrix



KLII-IH
Charged option



KL8II-2H
Charged option



KL8I3-2H
Charged option

Fire box



KL8I5H
Standard

Floor style



Checked steel plate

Load Classification

Load Types are Subject to Use Scenarios.

Type-A Load

It is ordinary load for loading and unloading goods manually or by cart. Goods should be distributed on car platform evenly and can't be piled at any special spot of the car platform.



Type-B Load

Dedicated car elevator load.

Type-C Load (C1 & C2)

C1 Load

C1: It is industrial transportation load, loading and unloading goods by cart or electric carrier.
The goods should be distributed on the car platform evenly and can't be piled at any special spot of the car platform.
The loader can't be transported by elevator.
During loading and unloading, the total weight of the carrier and goods should not exceed 60% of the rated load.



C2 Load

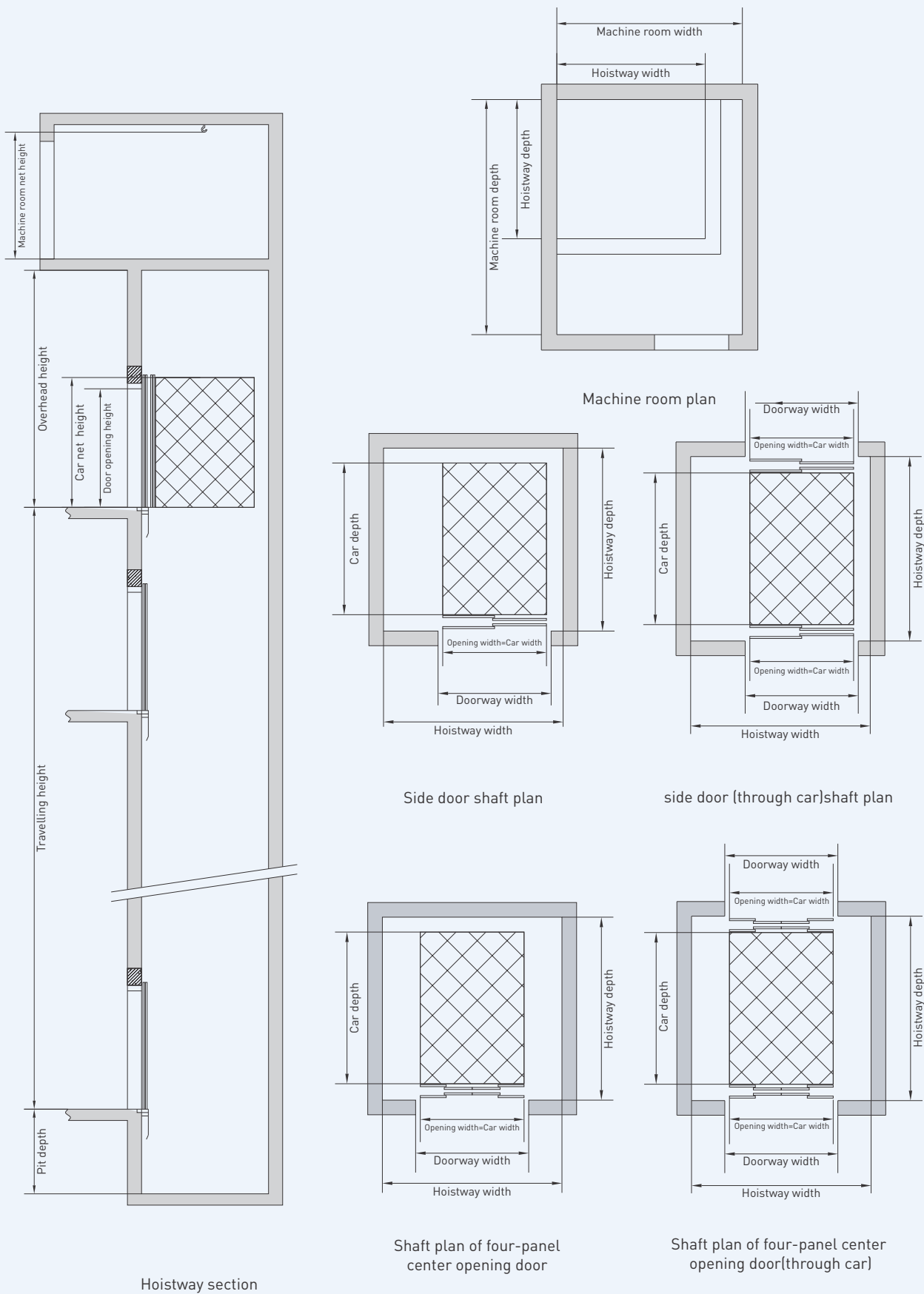
C2: It is industrial transportation load, loading and unloading goods by electric carrier or industrial forklift.
The loader can be transported by elevator.
During loading and unloading, total weight of the carrier and goods should not exceed 85% of the rated load.

Scope of Application

Model	Load	Standard load	Paid optional load
KLH Machine Room Freight Elevator	1000	A	-
	1600	A	-
	2000	A	C2
	3000	A,C1	C2
	4000	A,C1	C2
	5000	A,C1,C2	-
	6000	A,C1,C2	-
	7000	A,C1,C2	-
	8000	A,C1,C2	-
	9000	A,C1,C2	-
	10000	A,C1,C2	-
KLW Machine Room Less Freight Elevator	1600	A	-
	2000	A	C2
	3000	A,C1	C2
	4000	A,C1	C2
	5000	A,C1,C2	-
KLQ Car Elevator	3000	B	-
	4000		

Note: All dimensions are in millimeters (mm).

Freight Elevator (with Machine Room)



Geared Traction-type Freight Elevator

(Kg)	(m/s)	Traction	(mm)	(mm)	(mm) Car size			(mm) Hoistway size		(mm) Machine room		(mm) Overhead	(mm) Pit depth	(mtr)
Loading capacity	Rated speed	ratio	Door opening width	Door opening height	Width	Depth	Height	Width	Depth	Width	Depth	Height		Max.travelling height
6000	0.5	4.1	2600	2300	2600	4000	2500	4200	4500	4200	4500	4500	1900	44
	0.25											4600	2000	15
7000	0.5		2800	2300	2800	4300	2500	4550	4800	4550	5500	4600	2000	44
	0.25											4600	2000	15
8000	0.5		2800	2300	2800	4900	2500	4550	5400	4550	6100	4600	2000	44
	0.25											4600	2000	15
9000	0.5		3000	2300	3000	5000	2500	4750	5500	4750	6200	4600	2000	44
	0.25											4600	2000	15
10000	0.5		3000	2300	3000	5500	2500	4750	6000	4750	6500	4600	2000	44
	0.25											4600	2000	15

Remarks Car net height excluding ceiling

Explosion-proof Cargo Elevator

(Kg)	(m/s)	Traction	(mm)	(mm)	(mm) Car size			(mm) Hoistway size		(mm) Machine room		(mm) Overhead	(mm) Pit depth	(mtr)
Loading capacity	Rated speed	ratio	Door opening width	Door opening height	Width	Depth	Height	Width	Depth	Width	Depth	Height		Max.travelling height
2000	0.5	2.1	1500	2100	1500	2700	2300	2700	3200	3400	4500	4150	1450	44
3000	0.5		2200	2100	2200	2600	2300	3600	3100	3600	4400	4300	1500	44
4000	0.5		2200	2300	2200	3200	2500	3800	3700	3800	4100	4000	1500	44
5000	0.5	4.1	2400	2300	2400	3600	2500	4100	4100	4100	4100	4500	1500	44

Remarks Car net height excluding ceiling

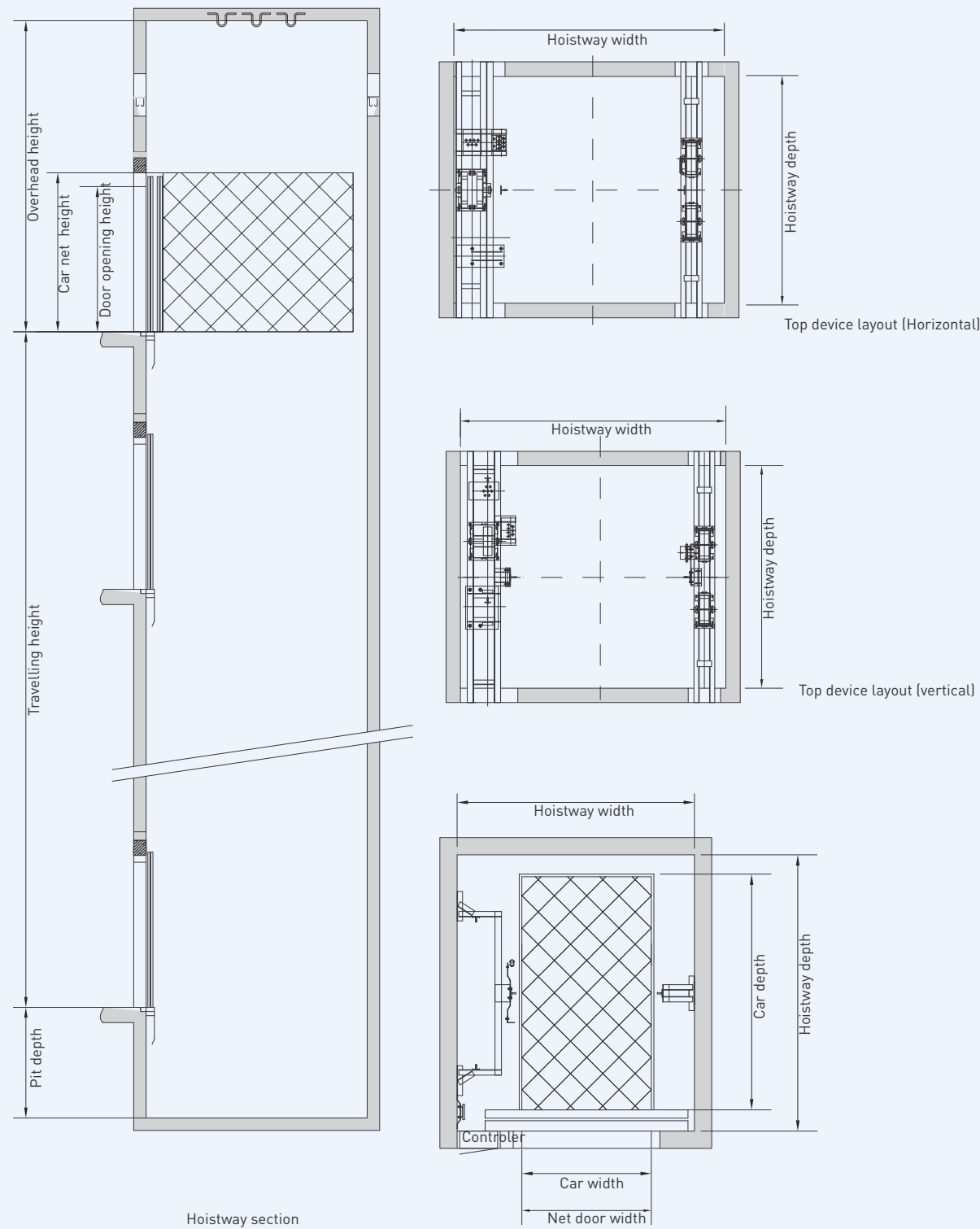
Gearless Traction-type Freight Elevator

(Kg)	(m/s)	Traction	(mm)	(mm)	(mm) Car size			(mm) Hoistway size		(mm) Machine room size		(mm) Overhead	(mm) Pit	(mtr)
Loading capacity	Rated speed	ratio	Door opening width	Door opening height	Width	Depth	Height	Width	Depth	Width	Depth	height	depth	Max.travelling height
1000	0.5	2.1	1300	2100	1300	1750	2300	2400	2200	2400	2200	3950	1300	50
	1											3950	1300	50
	1.6											4150	1450	60
	1.75											4250	1450	90
	2											4400	1500	105
1600	0.5	2.1	1500	2100	1500	2250	2300	2700	2750	3400	4500	4250	1450	30
	1											4250	1450	50
	1.6											4400	1550	80
	1.75											4550	1600	90
	2											4700	1650	90
2000	0.5	2.1	1500	2100	1500	2700	2300	2700	3200	3400	4500	4250	1450	30
	1											4250	1450	50
	1.6											4400	1550	80
	1.75											4550	1600	90
	2											4700	1650	90
2000	0.5	2.1	1500	2100	1500	2700	2300	2700	3200	3400	4500	4050	1450	28
	0.63											4400	1450	38
	1											4400	1450	76
3000	0.5	2.1	2200	2100	2200	2600	2300	3600	3100	3600	4400	4050	1500	28
	0.63											4500	1500	38
	1											4500	1500	76
4000	0.5	4.1	2200	2300	2200	3200	2500	3900	3700	3900	3700	4400	1550	28
	0.63											4400	1550	38
	1											4400	1550	60
5000	0.5	4.1	2400	2300	2400	3600	2500	4100	4100	4100	4100	4500	1550	28
	0.63											4500	1600	38
	1											4500	1600	60

Remarks Car net height excluding ceiling

Note: All dimensions are in millimeters (mm).

Freight Elevator (Machine Roomless)



Machine Roomless Freight Elevator Parameter Table (Horizontal)

(Kg)	(m/s)	Traction	(mm)	(mm)	(mm)			(mm)		(mm)	(mm)	(mtr)
					Car size			Hoistway size		Overhead	Pit	Max.travelling
Loading capacity	Rated speed	ratio	Door opening width	Door opening height	Width	Depth	Height	Width	Depth	height	depth	height
1600	0.5	2.1	1500	2100	1500	2250	2300	2600	2650	4150	1400	30
	1.0									4200	1400	40
	1.6									4350	1500	50
2000	0.5		1500	2100	1500	2700	2300	2750	3200	4500	1500	30
	0.63									4500	1500	38
	1.0									4500	1500	60
3000	0.5		2200	2100	2200	2600	2300	3700	3100	4500	1500	30
	0.63									4500	1500	38
	1.0									4500	1500	60
3000	0.5	4:1	2000	2100	2000	2900	2300	3500	3400	4500	1500	30
	0.63									4500	1500	38
	1.0									4500	1500	60
4000	0.5		2200	2300	2200	3200	2500	3800	3700	4800	1800	30
	0.63									4800	1800	38
	1.0									5000	2100	60
5000	0.5		2400	2300	2400	3600	2500	4100	4100	4800	1800	30
	0.63									4800	1800	38
	1.0									5000	2100	60
Remarks												

2000, 3000, 4000kg are 4:1 structure Car net height excluding ceiling

Machine Roomless Freight Elevator Parameter Table (Vertical)

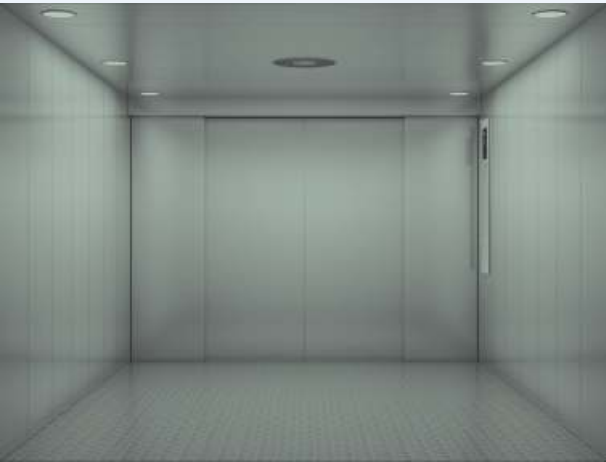
(Kg)	(m/s)	Traction	(mm)	(mm)	(mm)			(mm)		(mm)	(mm)	(mtr)
					Car size			Hoistway size		Overhead	Pit	Max.travelling
Loading capacity	Rated speed		Door opening width	Door opening height	Width	Depth	Height	Width	Depth	height	depth	height
2000	0.5		1500	2100	1500	2700	2300	2800	3200	4200	1600	30
	1.0									4200	1600	60
3000	0.5		2200	2100	2200	2600	2300	3700	3100	4300	1600	30
	1.0									4300	1600	60
4000	0.5	4.1	2200	2300	2200	3200	2500	3900	3700	4800	2000	30
	0.63									4800	2000	38
	1.0									4800	2000	60
5000	0.5		2400	2300	2400	3600	2500	4100	4100	4800	2000	30
	0.63									4800	2000	38
	1.0									4800	2000	60

Note: All dimensions are in millimeters (mm).

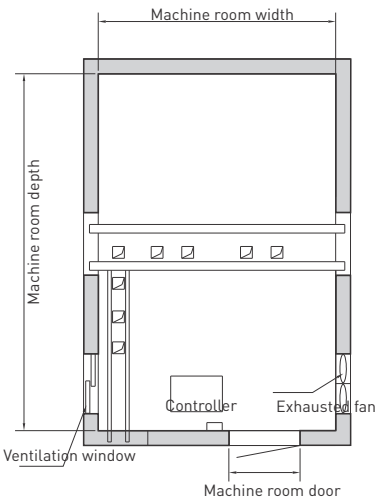
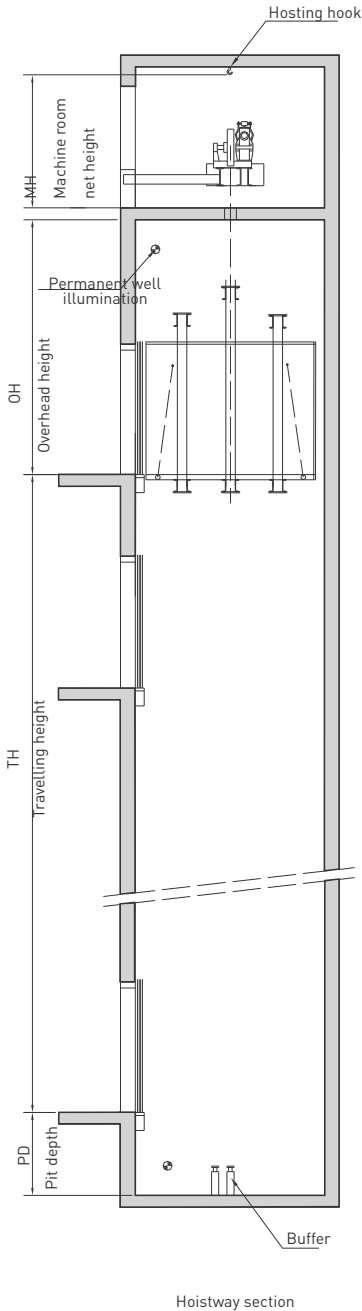
With the perfect car depth and the best operation panel, elevator allows various cars to get in and out of it easily. The efficient combination of mature car lifting technology and great technological strength optimizes smooth and noise-free product characteristics of car elevator. It is an ideal choice for parking and vertical transportation in many scenarios.



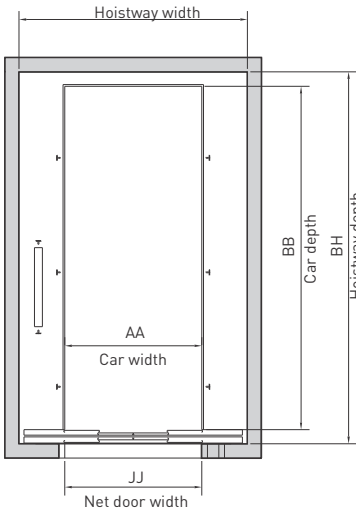
Standard configuration of the lift car decoration



- Car ceiling**
Dusting steel plate
- Ventilation**
Round fan
- Car wall**
Dusting steel plate
- Car door**
Dusting steel plate
- Floor**
Checked steel plate
- Cop.**
KL1113-4C
- Hop**
KL1213-1H



Machine room plan



Four-panel center opening door layout

Car Elevator Parameter Table

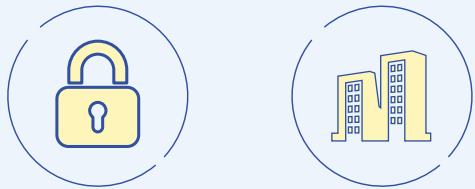
(Kg)	(m/s)	Traction	(mm) Door opening	(mm) Door opening	(mm) Car size			(mm) Hoistway size		(mm) Machine room size		(mm) Overhead	(mm) Pit	(mtr) Max.travelling
Loading capacity	Rated speed	ratio	width	height	Width	Depth	Height	Width	Depth	Width	Depth	height	depth	height
3000	0.5	4:1	2400	2300	2400	6000	2500	4000	6500	4000	6500	4500	1500	44
4000			3000	2300	3000	6000	2500	4700	6500	4700	6500	4500	1500	44

Note: All dimensions are in millimeters (mm).

Dumbwaiter Elevator

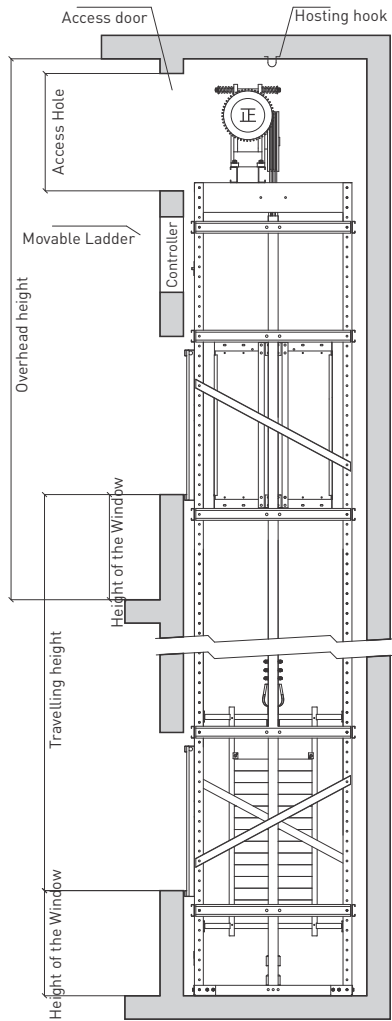


The core control system operating reliably collocated with the integrated landing door lock, and car door braking device and door protection system with super high security coefficient, offers multiple safety and protection, which can perfectly present the first level safety system of Dumbwaiter Elevator

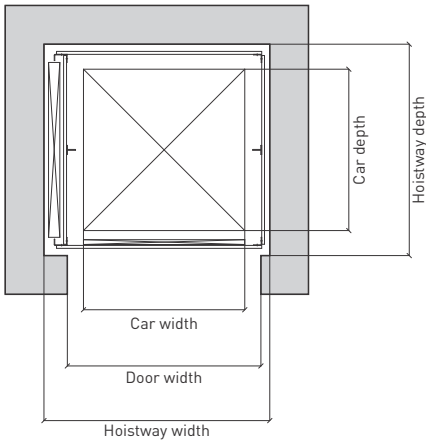


Multiple Installation Modes
Floor type, window type and table-top type

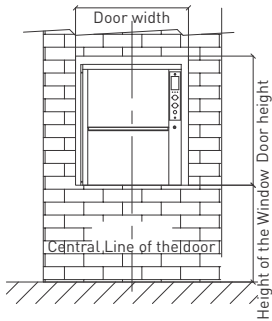
Multiple Types of Door Opening
Manual door, opposite door and vertical center opening door



Hoistway section



Hoistway plan



Door arrangement

Dumbwaiter Elevator Parameter Table

Loading capacity	(Kg)	(m/s)	(mm) Car size			(mm) Hoistway size		(mm) Clear door size		(mm) Overhead	(mm) Pit	(mtr) Max.travelling height
			Width	Depth	Height	Width	Depth	Width	Depth	height	depth	height
100		0.4	600	700	800	1000	1000	590	780	3200	/	25
		1	600	700	800	1000	1000	590	780	3200	/	45
200		0.4	800	800	900	1200	1100	790	880	3400	/	25
		1	800	800	900	1200	1100	790	880	3400	/	45
250		0.4	1000	1000	1100	1400	1235	980	1050	3500	500	25
		1	1000	1000	1100	1400	1235	980	1050	3500	500	45
300		0.4	1000	1000	1200	1450	1235	980	1150	3500	650	25
		1	1000	1000	1200	1450	1235	980	1150	3500	650	45

Note: The floor type is optional

Note: All dimensions are in millimeters (mm).

Standard Functional Feature

Brake detection protection	Brake arm detection switch executes real-time monitor to brake opening / closing. If the brake does not open by the requirements, the system will not start the elevator.
Protection of none door opening outside door zone	Automatic door opening is forbidden under none door zone condition.
Low-speed protection	It prevents elevator from low-speed running outside the control range causing any safety problem(s).
Prompt of overload alarm	Before the elevator shuts the door, if it detects overload signal, the elevator doors keep opening and overload buzzer rings and overload light is on. After it removes overload problem, it re-starts automatically.
Automatic parking in trouble	When the elevator stops in none leveling zone due to any trouble, if it is not caused by safety device action, it executes self-rescue running towards the middle floor direction at low speed. It parks and opens the door at the nearest service floor to prevent from any trapping accident.
Overspeed protection	With the star-delta brake function, the elevator system can limit the speed well even in case of forced brake releasing.
Unintended car movement protection (UCMP)	When the landing door isn't locked and the car door isn't closed, the elevator is equipped with a device which can stop unintended movements of the car leaving the landing caused by any failed part of the traction machine or drive control system for the safe operation of the car.
Checking the short circuit of door locks	It will check whether the landing door lock or car door lock has a short circuit every time when the elevator stops at a landing and opens the door. If any, the elevator will report the failure and stop until the failure is removed.
Fire-fighting return	When the fire-fighting switch closes, the system enters into fire-fighting return. The system will remove all the landing calls and car command signals, and it automatically returns to fire-fighting home landing, and the door keeps open. If the elevator is operating in reverse direction, it will stop at the nearest storey but won't open the door, then back to fire station and no longer running, the door will keep opening.
Home landing elevator locking	When the elevator is under automatic condition, when it shuts home landing locking switch, the elevator will complete all the registered car command services, automatically return to home landing, it enters into parking condition. As soon as locking switch is on, it immediately returns from parking condition and goes into normal running.
Cancellation of the error register	If the passenger presses the car command button at certain floor and has registered it by mistake, (by then the car indicating lamp is bright), if under the condition of none elevator running, the passenger only double-click the car command button at the same floor to cancel the register. (By then the car indicating lamp out.)
Automatic correction of floor signal	If the floor display position has error, after the system detects terminal switch in the travel, it will automatically correct the floor display.
Cancellation of the reverse command(s) in the car	When the elevator travels to the farthest floor and changes the speed, it removes all the car command register.
Self-diagnostic embedded call button	If on certain storey, the call button was pressed over certain time but not disconnected, the system assumes that it is embedded button. It does not register the landing call of the same floor afterwards. Responsive lamp which corresponds to the button keeps flickering for alarming. When the button disconnects, it exits from above condition.
Re-leveling function	If the elevator parks on the current floor, when too many passengers enter into / leave the elevator, the car is separated from the leveling zone because of wire rope extension or shortening. The elevator will then travel to the leveling position at low speed under the condition of door-opening.

Healthy Elevator Ride (Optional)

Car UV Disinfection

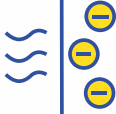


After the elevator has been unused for a certain period of time, it communicates and controls with the elevator to automatically light up the ultraviolet lamp for disinfection and sterilization. Ultraviolet rays can disinfect and sterilize various common pathogenic bacteria, with a sterilization rate of more than 99%, providing passengers with a safe and clean elevator environment.





Negative Ion Fresh Air System



Automatically releasing negative ions, adsorbing small dust particles, oxidizing foreign objects in the air, purifying the air, and caring for human health.

Car Air Purifier



Automatic dust removal and sterilization, clean and pollution-free, keeping the air inside the elevator car clean and hygienic at all times. The automatic operation time can be set, which is energy-saving and environmentally friendly.





📍 Cité Mourtada 2 Sipres Bande,
verte lot 104, VDN, Dakar 25000,
Senegal, West Africa.

✉ marketing@grvertiq.com

🌐 www.grvertiq.com

☎ +221 711 200 277